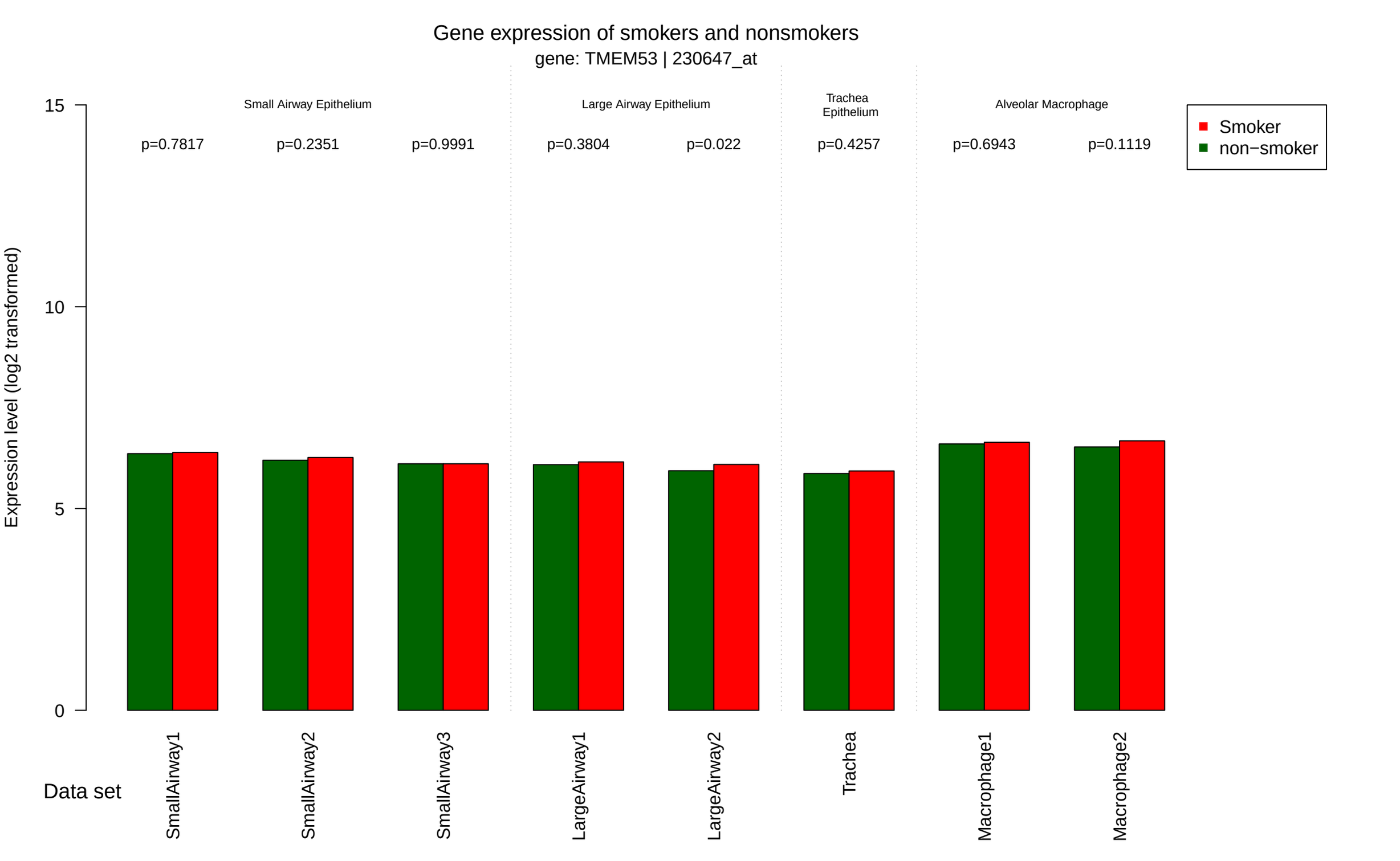
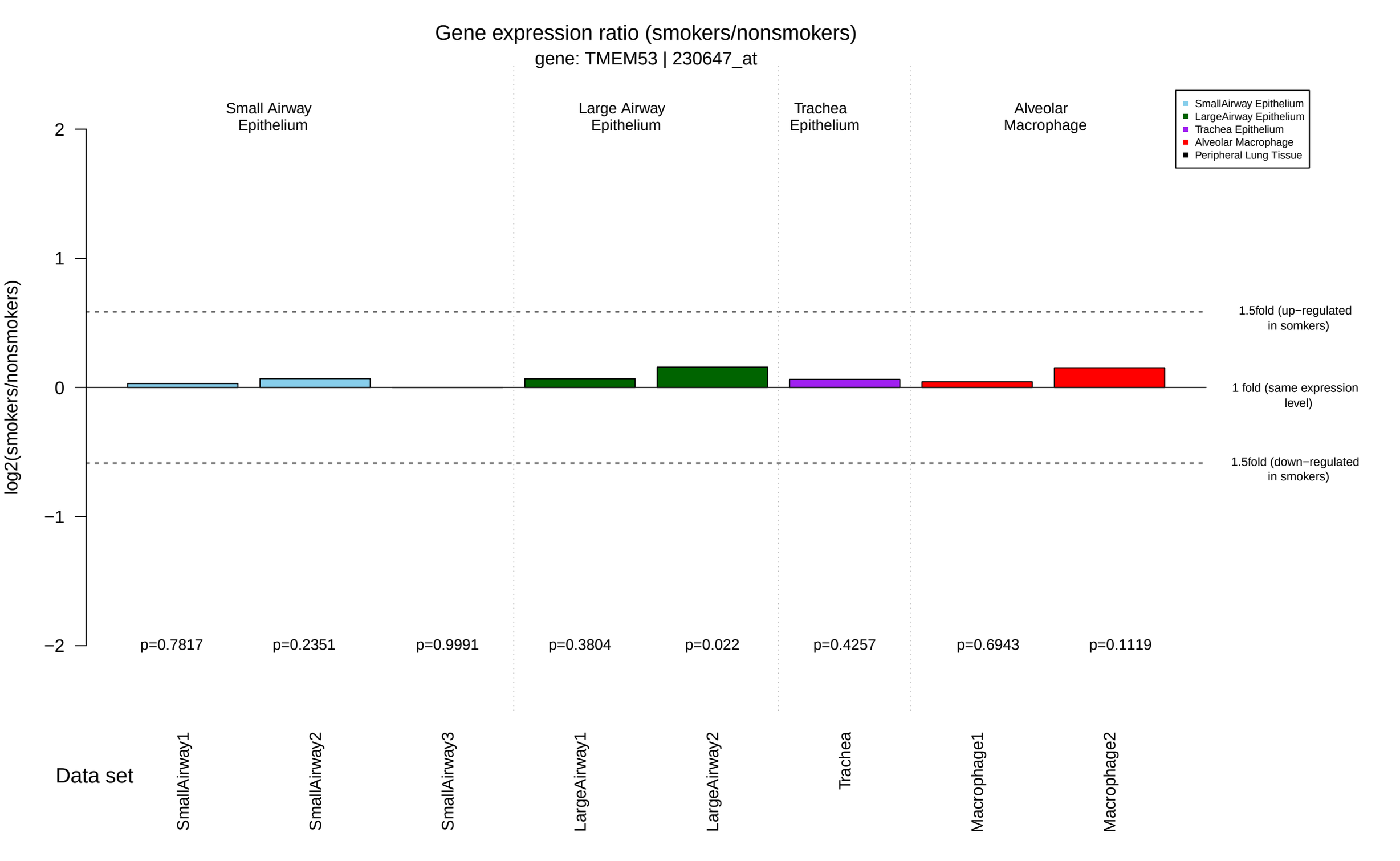
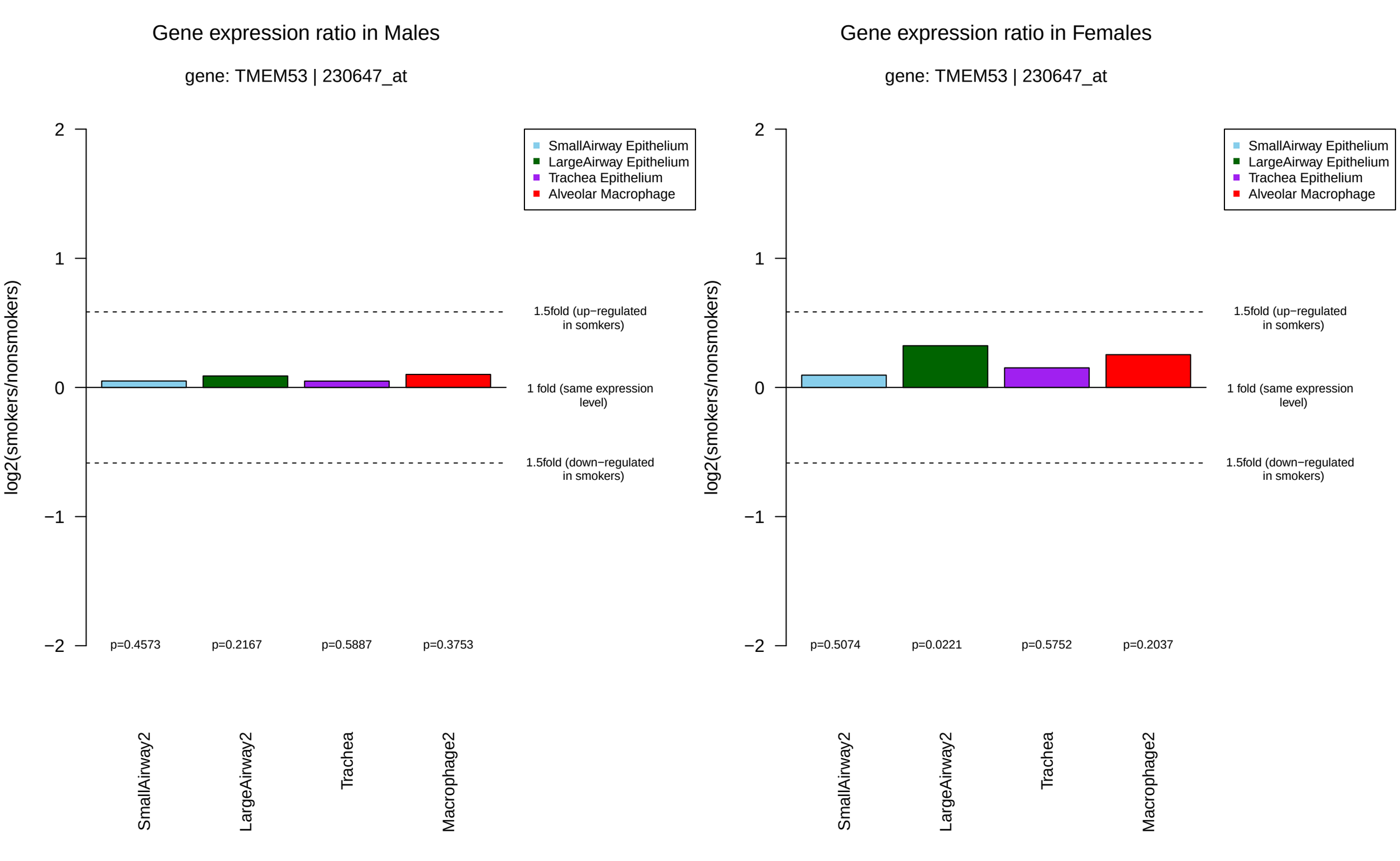
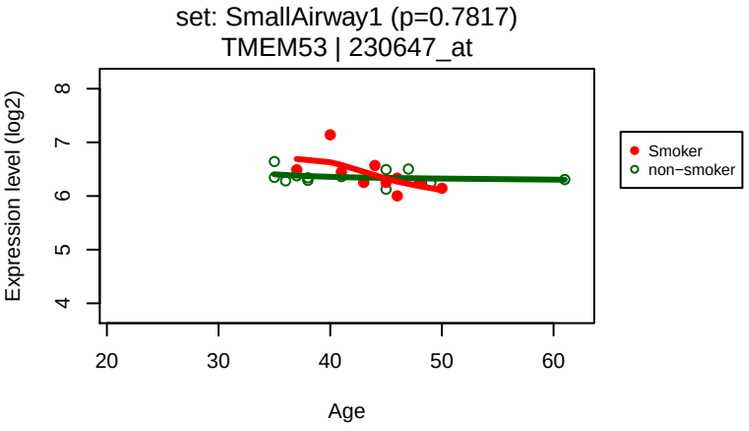
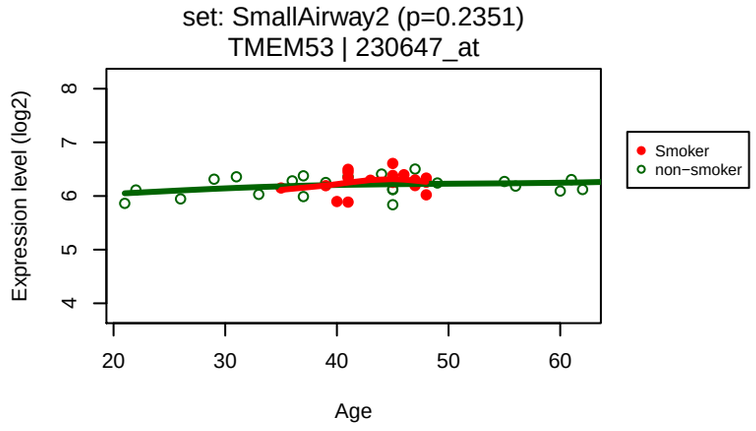
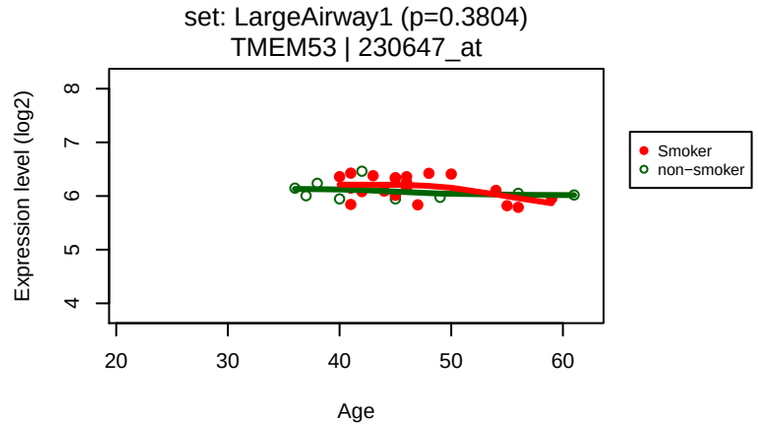

Gene expression of smokers and nonsmokers
gene: TMEM53 | 230647_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7817
p=0.2351
p=0.9991
p=0.3804
p=0.022
p=0.4257
p=0.6943
p=0.1119
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

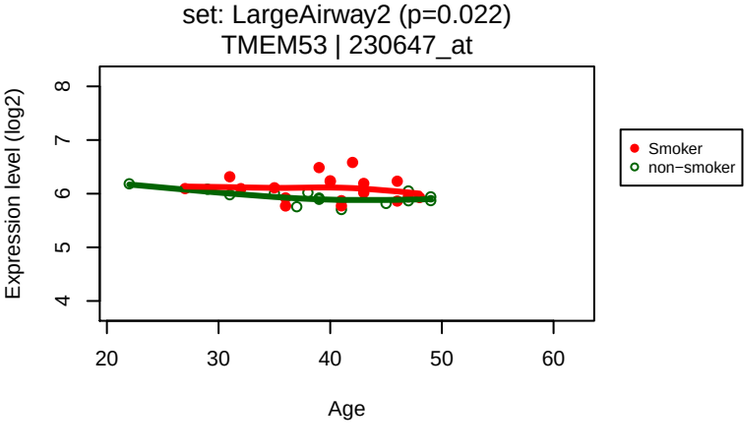

Gene expression ratio (smokers/nonsmokers)
gene: TMEM53 | 230647_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7817
p=0.2351
p=0.9991
p=0.3804
p=0.022
p=0.4257
p=0.6943
p=0.1119
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

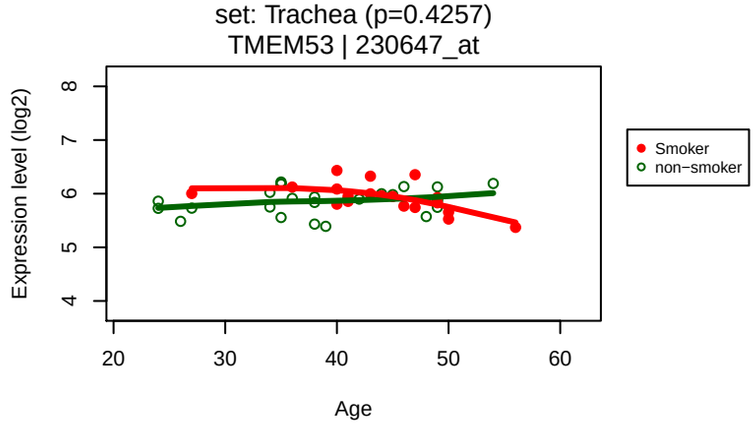

Gene expression ratio in Males
gene: TMEM53 | 230647_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4573
p=0.2167
p=0.5887
p=0.3753
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TMEM53 | 230647_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5074
p=0.0221
p=0.5752
p=0.2037
SmallAirway2
LargeAirway2
Trachea
Macrophage2

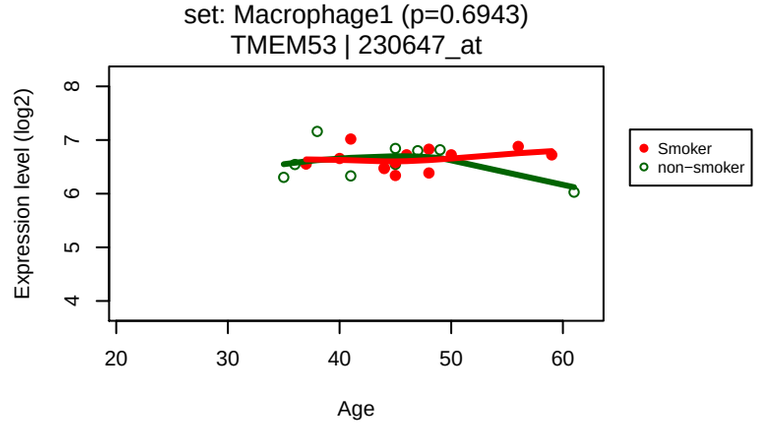

set: SmallAirway1 (p=0.7817)
TMEM53 | 230647_at
Expression level (log2)
Age
Smoker
non−smoker

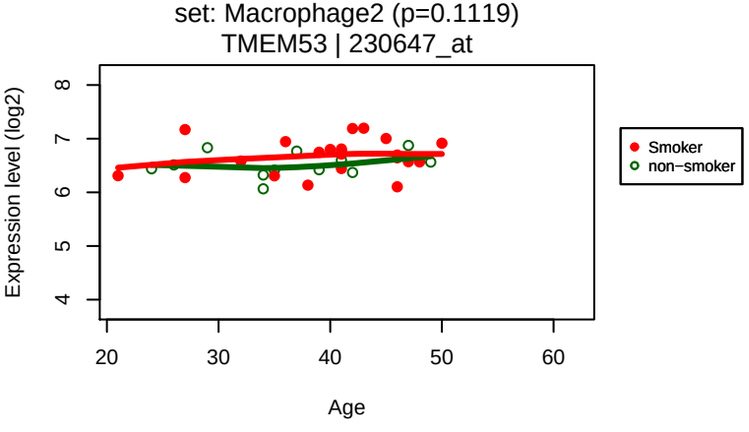

set: SmallAirway2 (p=0.2351)
TMEM53 | 230647_at
Expression level (log2)
Age
Smoker
non−smoker

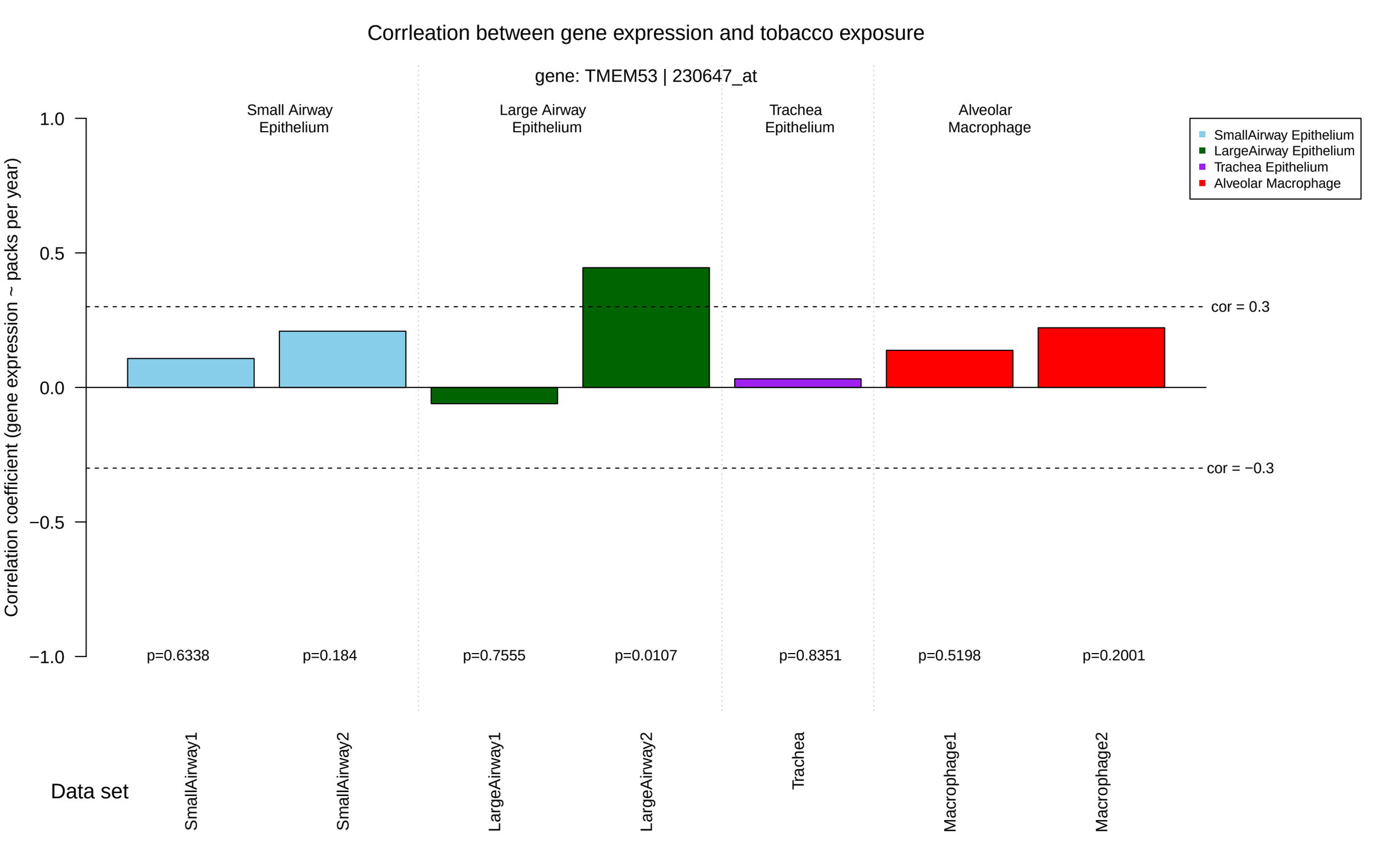

set: LargeAirway1 (p=0.3804)
TMEM53 | 230647_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.022)
TMEM53 | 230647_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4257)
TMEM53 | 230647_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6943)
TMEM53 | 230647_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1119)
TMEM53 | 230647_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TMEM53 | 230647_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6338
p=0.184
p=0.7555
p=0.0107
p=0.8351
p=0.5198
p=0.2001
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2